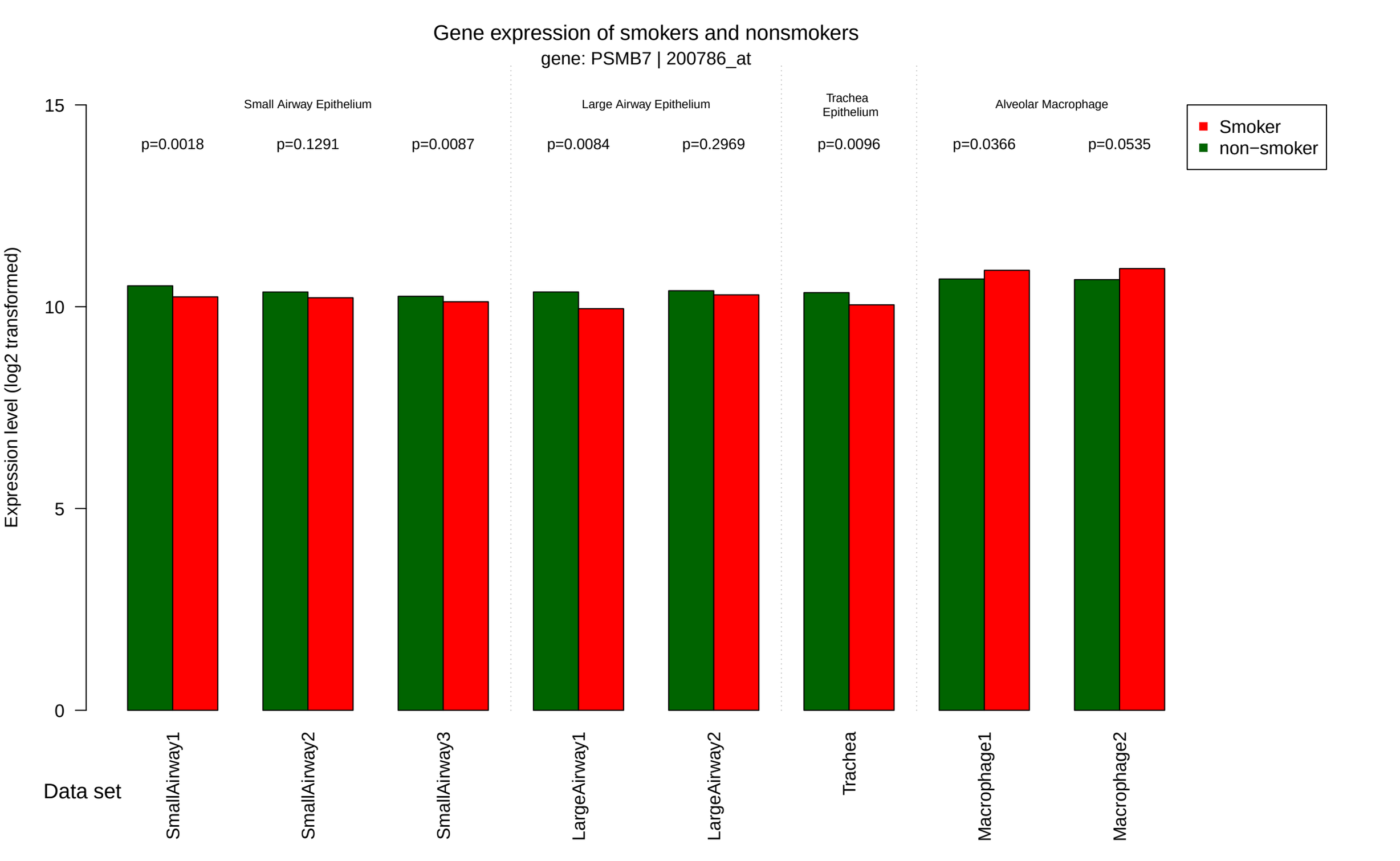
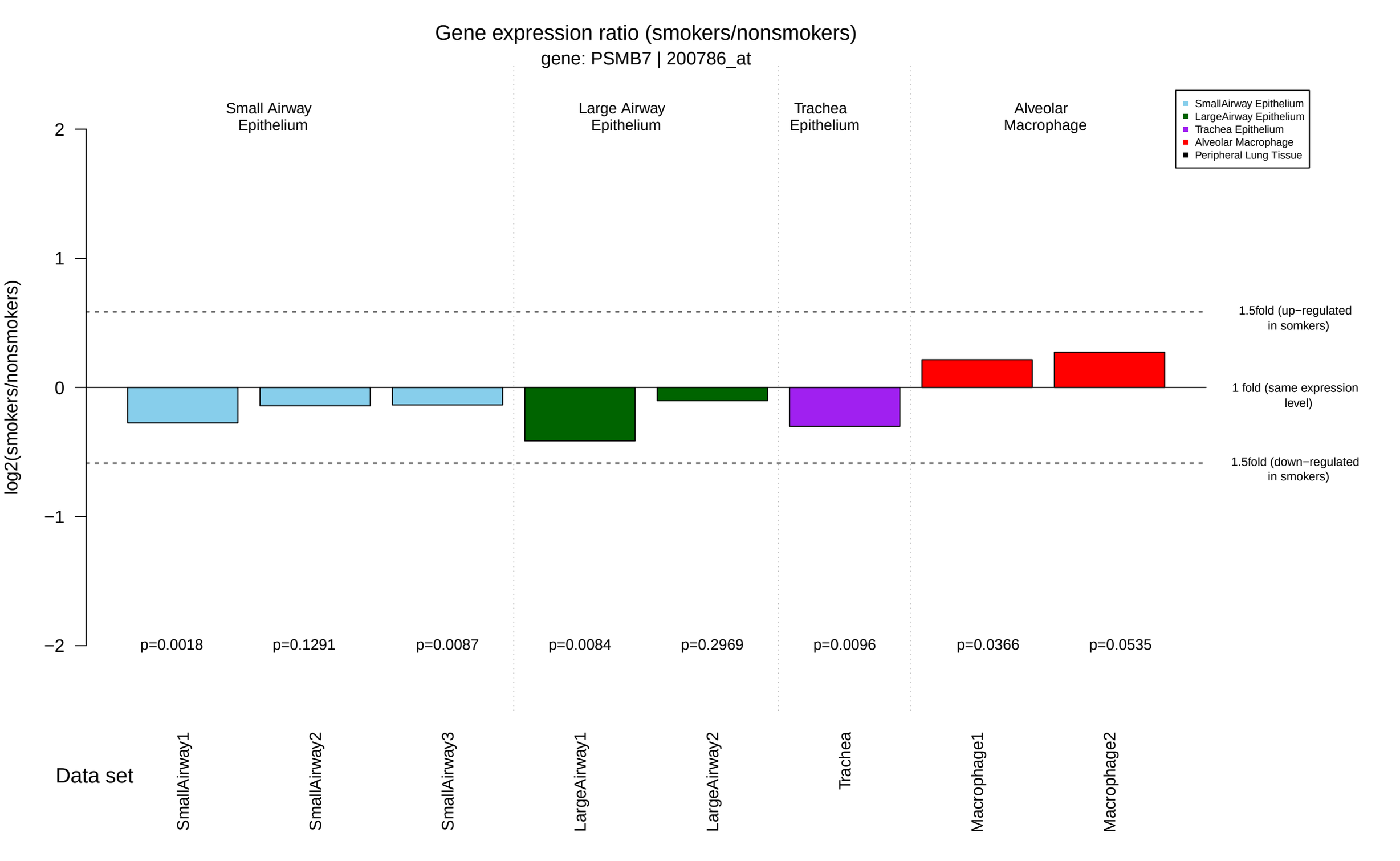
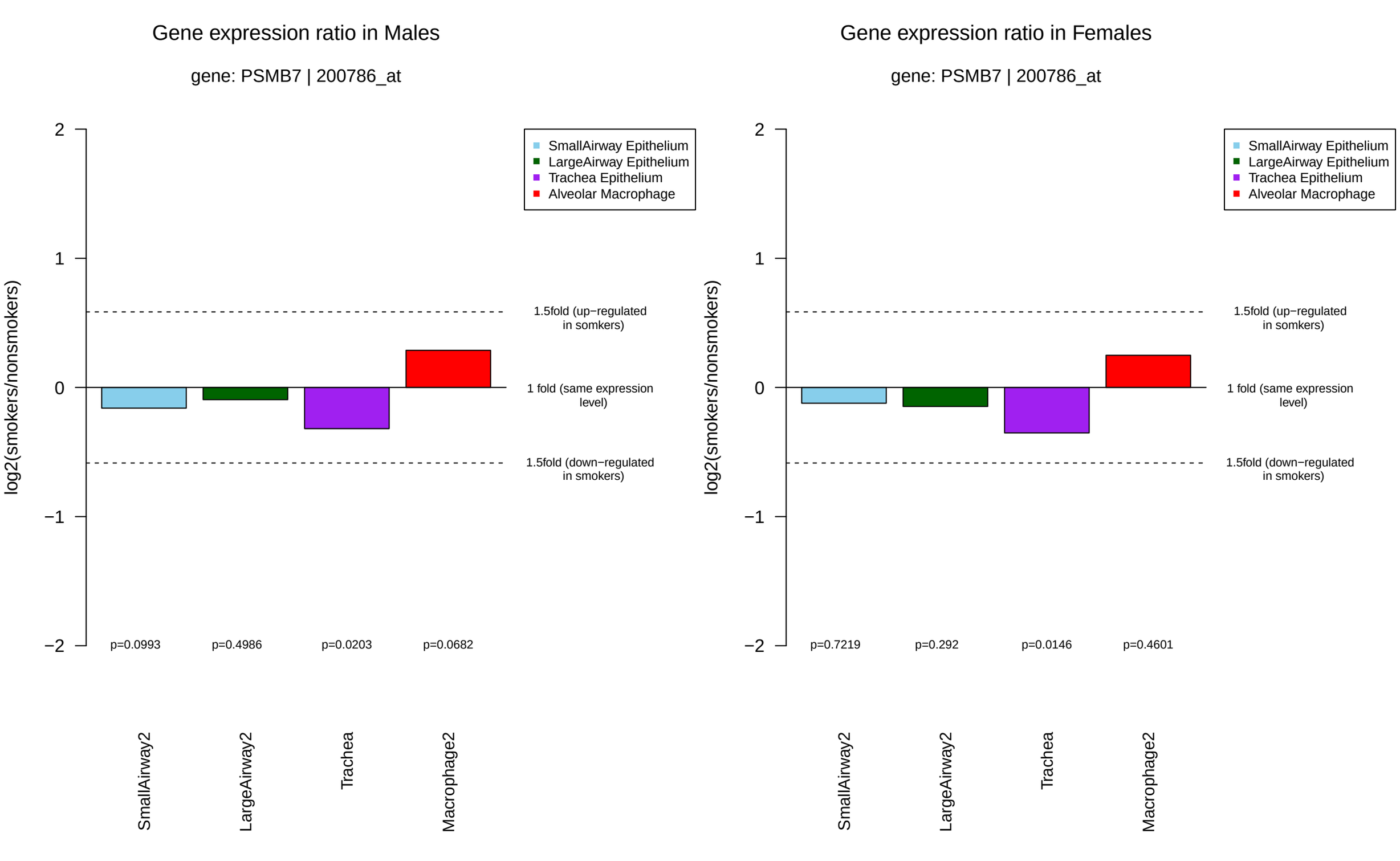
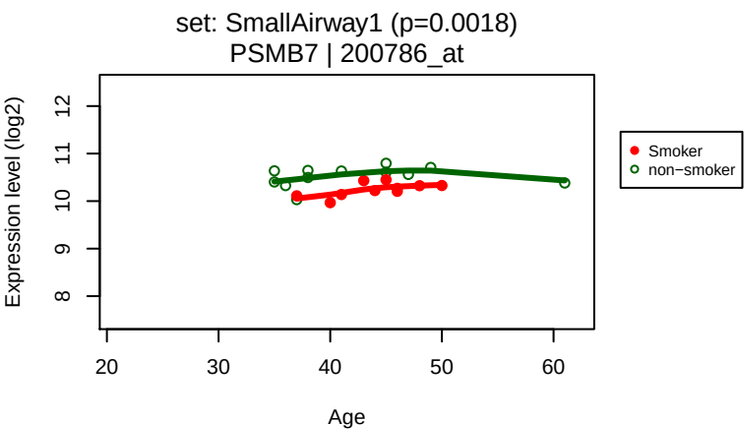
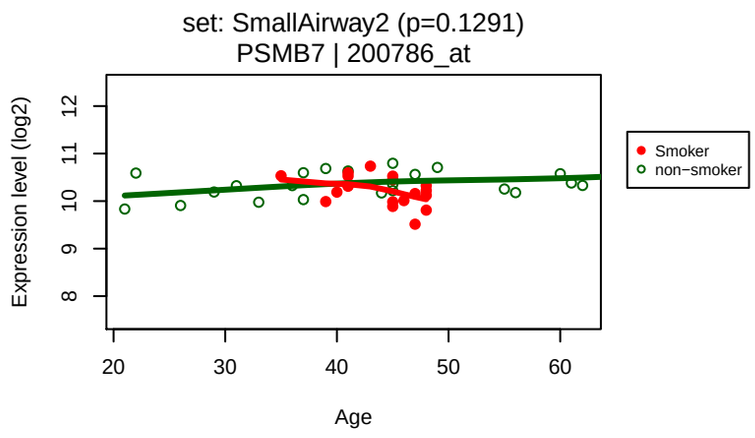
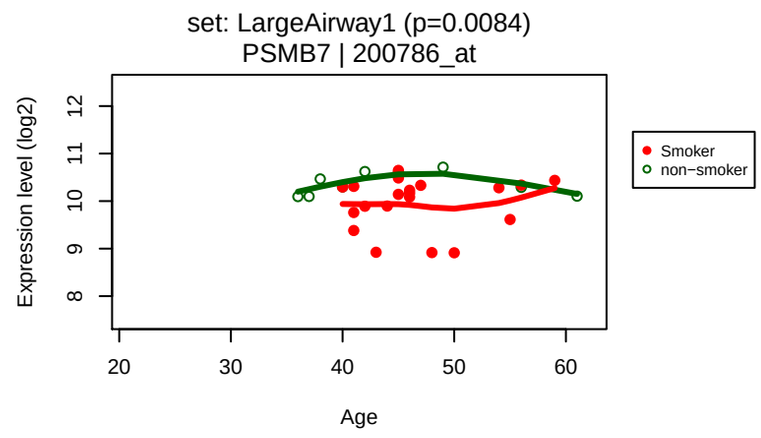

Gene expression of smokers and nonsmokers
gene: PSMB7 | 200786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0018
p=0.1291
p=0.0087
p=0.0084
p=0.2969
p=0.0096
p=0.0366
p=0.0535
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

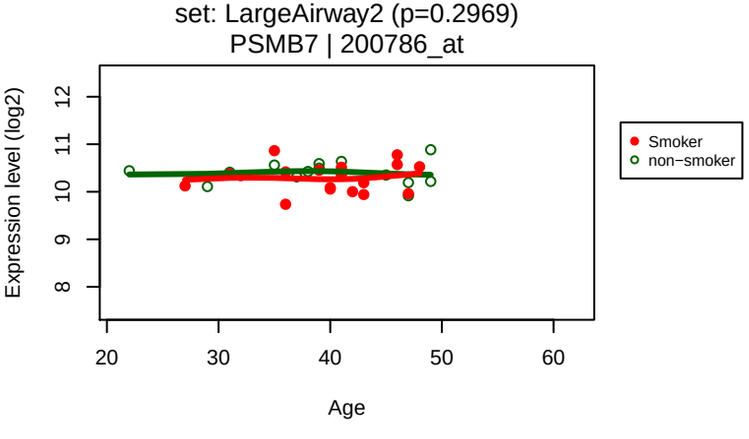

Gene expression ratio (smokers/nonsmokers)
gene: PSMB7 | 200786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0018
p=0.1291
p=0.0087
p=0.0084
p=0.2969
p=0.0096
p=0.0366
p=0.0535
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

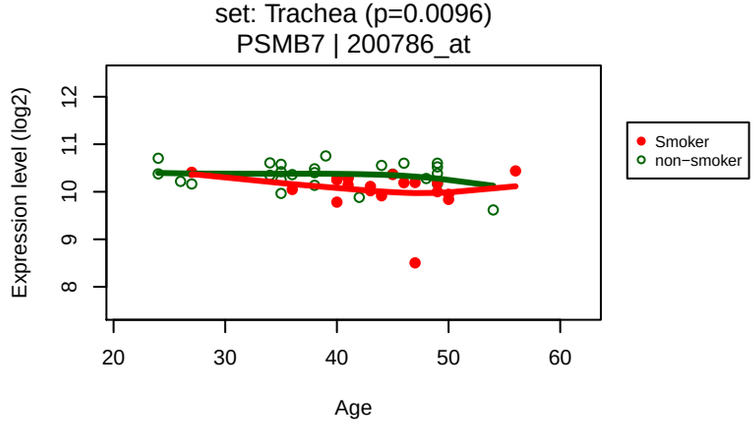

Gene expression ratio in Males
gene: PSMB7 | 200786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0993
p=0.4986
p=0.0203
p=0.0682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB7 | 200786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7219
p=0.292
p=0.0146
p=0.4601
SmallAirway2
LargeAirway2
Trachea
Macrophage2

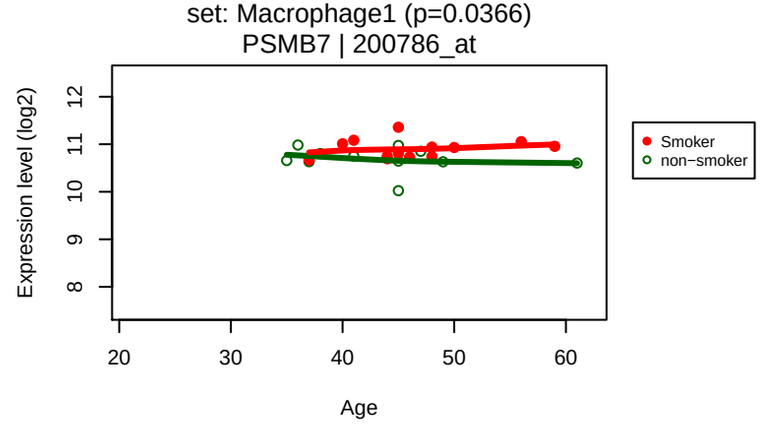

set: SmallAirway1 (p=0.0018)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker

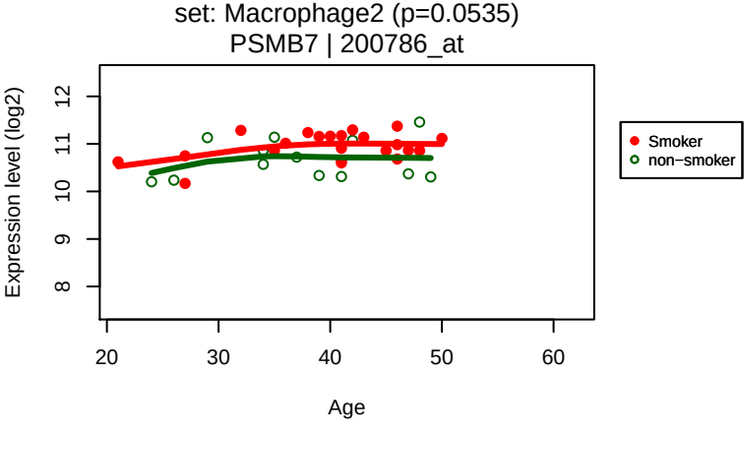

set: SmallAirway2 (p=0.1291)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker

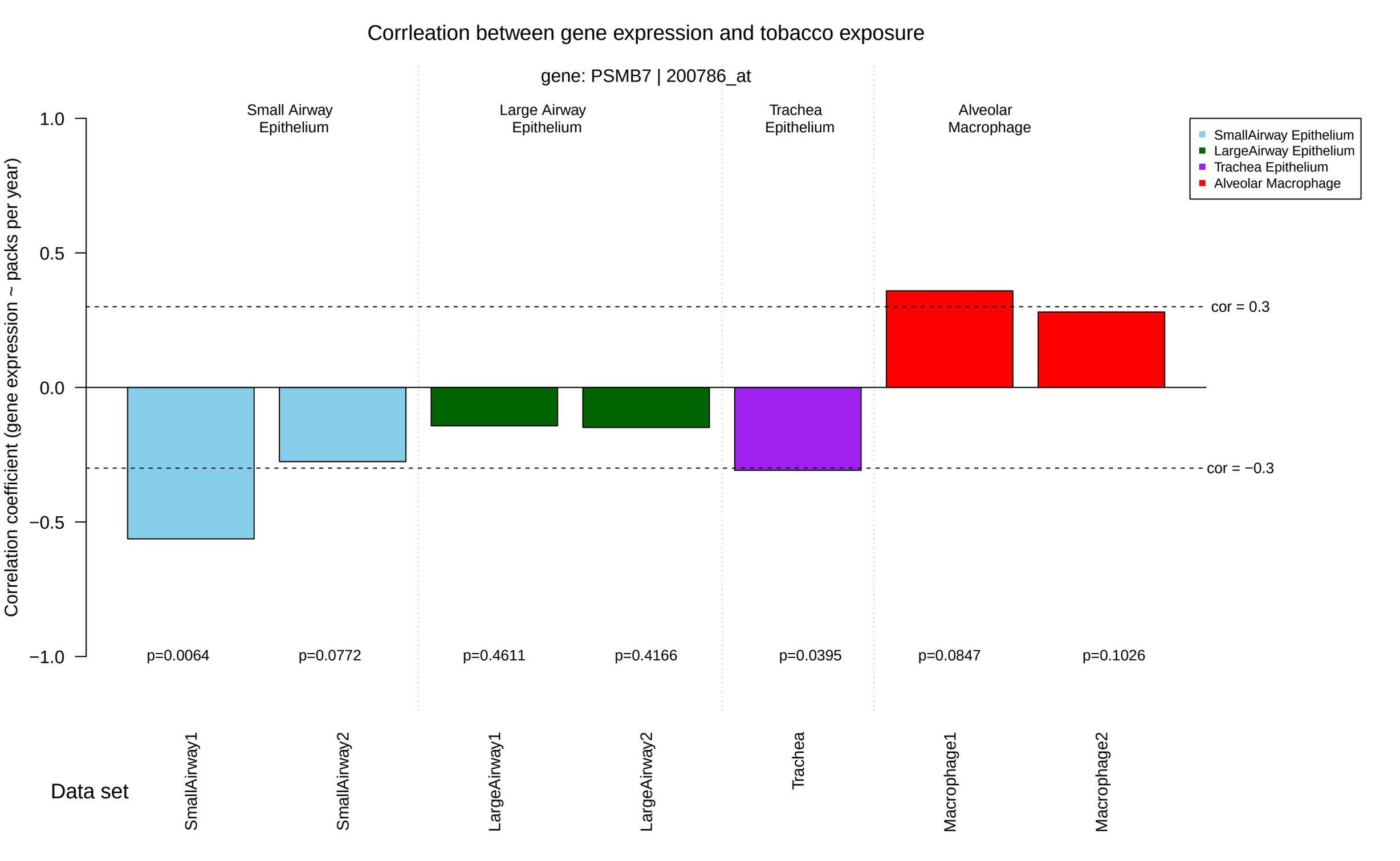

set: LargeAirway1 (p=0.0084)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2969)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0096)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0366)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0535)
PSMB7 | 200786_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB7 | 200786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0064
p=0.0772
p=0.4611
p=0.4166
p=0.0395
p=0.0847
p=0.1026
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2